# NACA 2412 Airfoil Aerodynamic Analysis

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## Abstract

The NACA 2412 airfoil was analysed at Reynolds numbers of 1, 3, 6 and 9 million (Mach 0.2) using XFLR5, which runs the XFoil panel method coupled with a viscous boundary-layer model. Lift, drag, pitching moment and surface pressure distributions were computed for angles of attack from −5° to 20°, with results post-processed in MATLAB and Python.

In the linear range the lift slope is 0.114 per degree and the zero-lift angle is −2.2°, independent of Reynolds number. Reynolds number mainly affects stall and drag: maximum lift rises from 1.53 at Re = 1 million to 1.88 at Re = 9 million, and the best lift-to-drag ratio rises from 101 to 136. Compared with NACA wind-tunnel data, the simulation matches the zero-lift angle and moment closely, but overpredicts maximum lift by about 7% and underpredicts minimum drag by about 13%, consistent with XFoil's smooth-surface and simplified-separation assumptions.

A comparison with the symmetric NACA 0012, run with identical settings, isolates the effect of camber. Camber raises lift by about 0.25 at every angle without changing the lift slope, cuts drag at cruise lift by 13 to 28%, and raises the best lift-to-drag ratio by 11 to 35%, at the cost of a constant nose-down moment of −0.053.

## 1. Introduction and objectives

An airfoil is the cross-section of a wing, and its shape decides how much lift and drag the wing produces. The NACA 2412 is a classic general-aviation section, used on aircraft such as the Cessna 152 and 172, and its wind-tunnel data is widely published. That makes it a good case for learning airfoil simulation and checking it against experiment.

The objectives of this project were to:

1. Simulate lift, drag and pitching moment of the NACA 2412 over a range of angles of attack.
2. Compute and interpret the surface pressure distribution.
3. Measure how Reynolds number affects performance, especially stall and drag.
4. Validate the simulation against NACA wind-tunnel data and explain any differences.
5. Compare the NACA 2412 with a second airfoil (Section 6).

# 2. Theory

## 2.1 Airfoil geometry

The NACA four-digit code describes the shape. For NACA 2412, maximum camber is 2% of the chord, located at 40% of the chord, and maximum thickness is 12% of the chord. All lengths are normalised so the chord c = 1.

## 2.2 Aerodynamic coefficients

Forces are expressed as dimensionless coefficients so that results apply to any wing size and speed. Per unit span, with air density ρ and freestream speed V:

$$C_l = \frac{L'}{\frac{1}{2}\rho V^2 c}, \qquad C_d = \frac{D'}{\frac{1}{2}\rho V^2 c}, \qquad C_m = \frac{M'_{c/4}}{\frac{1}{2}\rho V^2 c^2}$$

The pitching moment is taken about the quarter-chord point; a negative value means a nose-down moment.

## 2.3 Reynolds and Mach number

Reynolds number compares inertial and viscous forces, with μ the dynamic viscosity of air:

$$Re = \frac{\rho V c}{\mu}$$

Light aircraft wings operate at Re of a few million. Mach number M = V/a, with a the speed of sound. At M = 0.2 compressibility effects are small.

## 2.4 Pressure coefficient

The pressure coefficient compares local surface pressure p with freestream pressure p∞:

$$C_p = \frac{p - p_\infty}{\frac{1}{2}\rho V^2}$$

Cp = 1 at the stagnation point, where the flow comes to rest. Negative Cp is suction. The area enclosed between the upper and lower Cp curves gives the normal force, and so the lift.

## 2.5 Thin-airfoil theory

For a thin airfoil in inviscid flow, lift grows linearly with angle of attack α (in radians), where α(L=0) is the zero-lift angle:

$$C_l = 2\pi\left(\alpha - \alpha_{L=0}\right)$$

This gives a lift slope of 0.110 per degree. For NACA 2412 the theory predicts $\alpha_0 \approx -2.1°$.

## 2.6 Boundary layer, transition and stall

Near the surface, friction slows the air in a thin boundary layer. It starts laminar (smooth, low friction) and becomes turbulent (mixed, higher friction but more resistant to separation) at the transition point. At high angles the boundary layer cannot overcome the adverse pressure gradient behind the suction peak and separates from the upper surface. Lift then stops increasing: this is stall, and the peak lift is Cl,max.

## 2.7 How XFoil works

XFLR5 uses the XFoil solver, which couples two models:

1. **Inviscid panel method:** the surface is divided into straight panels with vortex singularities, and the flow is solved so that no air passes through the surface.
2. **Viscous boundary-layer model:** integral boundary-layer equations are solved along the surface, with transition predicted by the $e^n$ method. The two solutions are iterated until they agree.

The $e^n$ method's Ncrit parameter represents freestream turbulence. Ncrit = 9 corresponds to an average wind tunnel.

## 3. Methodology

The workflow ran from geometry to validated results in six steps:

1. **Geometry.** NACA 2412 coordinates were imported into XFLR5 (v6.62) as a Selig-format `.dat` file.
2. **Panel refinement.** The imported file had only 35 points, which gave a pointed leading edge and a jagged Cp curve. The geometry was regenerated with 200 panels, clustered at the leading and trailing edges.
3. **Analysis setup.** Type 1 (fixed Reynolds number) polars were defined with the settings below.
4. **Angle sweep.** Each polar was run from −5° to 20° in 0.5° steps, storing every operating point.
5. **Post-processing.** Polar data was extracted to CSV and plotted in MATLAB and Python. Lift slope and zero-lift angle were found by a linear fit over −4° to 6°.
6. **Verification and validation.** Cp distributions were integrated to check they reproduce the reported Cl, and results were compared with NACA wind-tunnel data.

| Setting | Value |
|---|---|
| Solver | XFLR5 v6.62 (XFoil) |
| Panels | 200 |
| Reynolds numbers | 1, 3, 6 and 9 million |
| Mach number | 0.2 |
| Ncrit | 9 (free transition) |
| Angle of attack | −5° to 20°, step 0.5° |

A few points did not converge: 1.5° at Re = 1 million, 5° and 5.5° at Re = 3 million, and 6.5° at Re = 6 million. These lie in regions where the transition point moves rapidly, and were left as gaps rather than forced.

## 4. Results

Reynolds number barely changes lift in the normal flight range but strongly affects stall and drag. Figure 1 shows the four polars and Table 1 lists the key values.

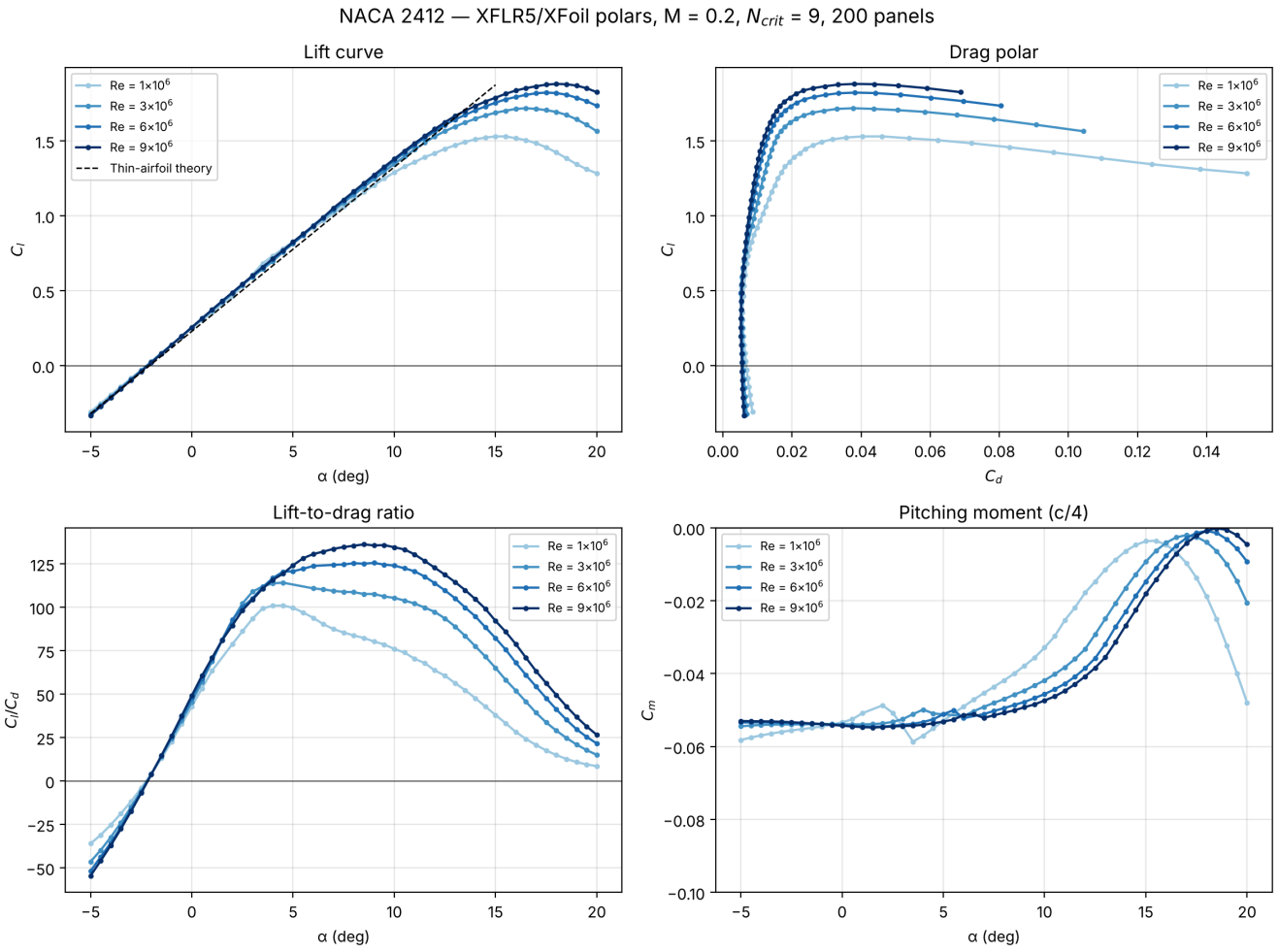



*Figure 1. Lift curve, drag polar, lift-to-drag ratio and quarter-chord moment, M = 0.2, Ncrit = 9.*

| Quantity | Re = 1M | Re = 3M | Re = 6M | Re = 9M |
|---|---|---|---|---|
| Lift slope (per degree) | 0.114 | 0.114 | 0.114 | 0.115 |
| Zero-lift angle (deg) | −2.2 | −2.2 | −2.2 | −2.2 |
| Cl,max | 1.53 | 1.72 | 1.82 | 1.88 |
| Stall angle (deg) | 15.5 | 16.5 | 17.5 | 18.0 |
| Minimum Cd | 0.0056 | 0.0052 | 0.0052 | 0.0052 |
| Best Cl/Cd | 101 at 4.5° | 114 at 4.5° | 126 at 9.0° | 136 at 8.5° |
| Cm, c/4 (−4° to 6° mean) | −0.054 | −0.053 | −0.053 | −0.054 |

*Table 1. Key performance values from XFLR5.*

## 4.1 Lift

The lift curve is linear up to about 10° and follows thin-airfoil theory closely. Cl at 0° is 0.25, because the camber produces lift even with no tilt. Above 10° the curves separate by Reynolds number: higher Re delays separation, so the airfoil reaches higher lift and stalls later.

## 4.2 Drag and efficiency

Minimum drag is about 0.0052, around Cl = 0.3 to 0.5. At high lift, drag rises much faster at Re = 1 million because the thicker, weaker boundary layer separates earlier. The best lift-to-drag ratio rises from 101 to 136 with Reynolds number, and its angle moves from 4.5° to about 9°.

## 4.3 Pitching moment

The quarter-chord moment stays nearly constant at about −0.054 up to roughly 10°, as expected for a cambered section. It becomes less negative approaching stall as the centre of pressure moves.

## 4.4 Pressure distribution

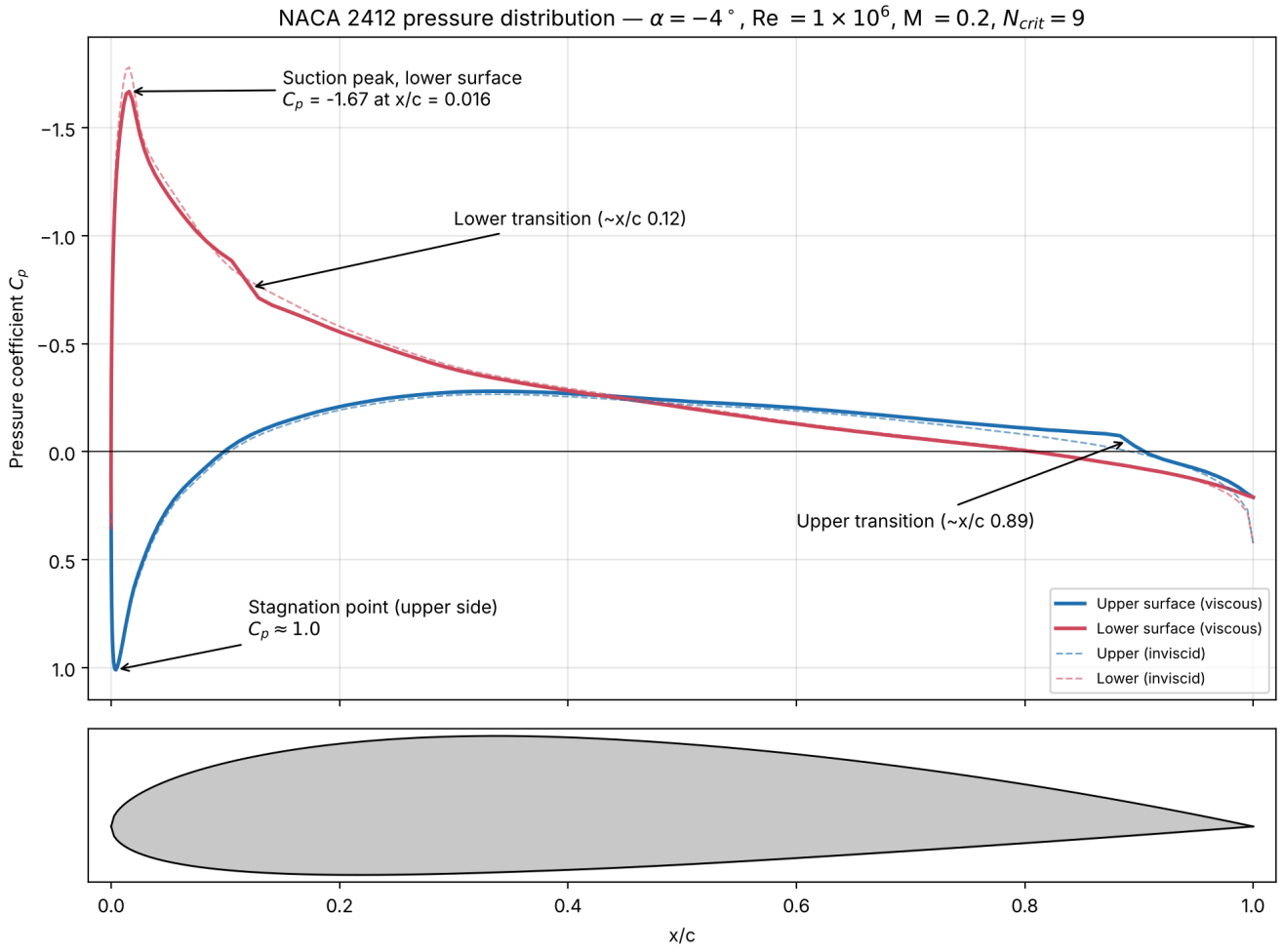


*Figure 2. Surface pressure distribution at α = −4°, Re = 1 million, M = 0.2.*

At −4° the stagnation point sits on the upper side of the nose and the suction peak (Cp = −1.67 at x/c = 0.016) is on the lower surface, giving negative lift. Transition appears as kinks at x/c ≈ 0.11 (lower) and 0.88 (upper). Integrating this distribution gives Cl = −0.196, matching XFLR5's reported −0.197, which verifies the pressure output.

*[To add: overlaid Cp plots at 0°, 5°, 10° and 17.5° at Re = 6 million, showing the growing suction peak and trailing-edge separation at stall.]*

## 5. Validation against wind-tunnel data

XFLR5 matches experiment closely for zero-lift angle, Cl at 0° and moment, but overpredicts lift slope and maximum lift and underpredicts drag. Table 2 compares the Re = 6 million results with NACA wind-tunnel data summarised by Abbott & von Doenhoff (1949).

| Quantity | XFLR5 (Re = 6M) | Experiment (approx.) | Difference |
|---|---|---|---|
| Cl at α = 0° | 0.255 | ≈ 0.25 | < 2% |
| Zero-lift angle (deg) | −2.2 | ≈ −2 | ≈ 0.2° |
| Lift slope (per degree) | 0.114 | ≈ 0.105 | ≈ +9% |
| Cl,max | 1.82 | ≈ 1.7 | ≈ +7% |
| Minimum Cd | 0.0052 | ≈ 0.006 | ≈ −13% |
| Cm, c/4 | −0.054 | ≈ −0.045 to −0.05 | small |

*Table 2. Simulation vs experiment. Experimental lift slope, Cd and Cm values are approximate and must be read from the NACA 2412 chart in Theory of Wing Sections before final submission.*

The differences have clear physical causes:

- **Lift slope too high.** XFoil models an ideal 2D section in unbounded flow and slightly underestimates boundary-layer displacement, which in reality reduces the effective camber and angle.
- **Cl,max too high.** Stall is driven by turbulent separation, which XFoil's integral boundary-layer model handles only approximately. It tends to keep the flow attached too long.
- **Drag too low.** XFoil assumes a perfectly smooth surface. Real models have small imperfections that trigger earlier transition and add skin friction.

The simulation is therefore reliable in the linear range (about −5° to 10°) and should be treated with caution near stall.

## 6. Comparison with the NACA 0012

Adding 2% camber shifts the whole lift curve up by about 0.25 without changing its slope, and makes the airfoil more efficient at every Reynolds number. The NACA 0012 has the same 12% thickness with no camber, so every difference below comes from camber alone. Both airfoils were run with identical settings.

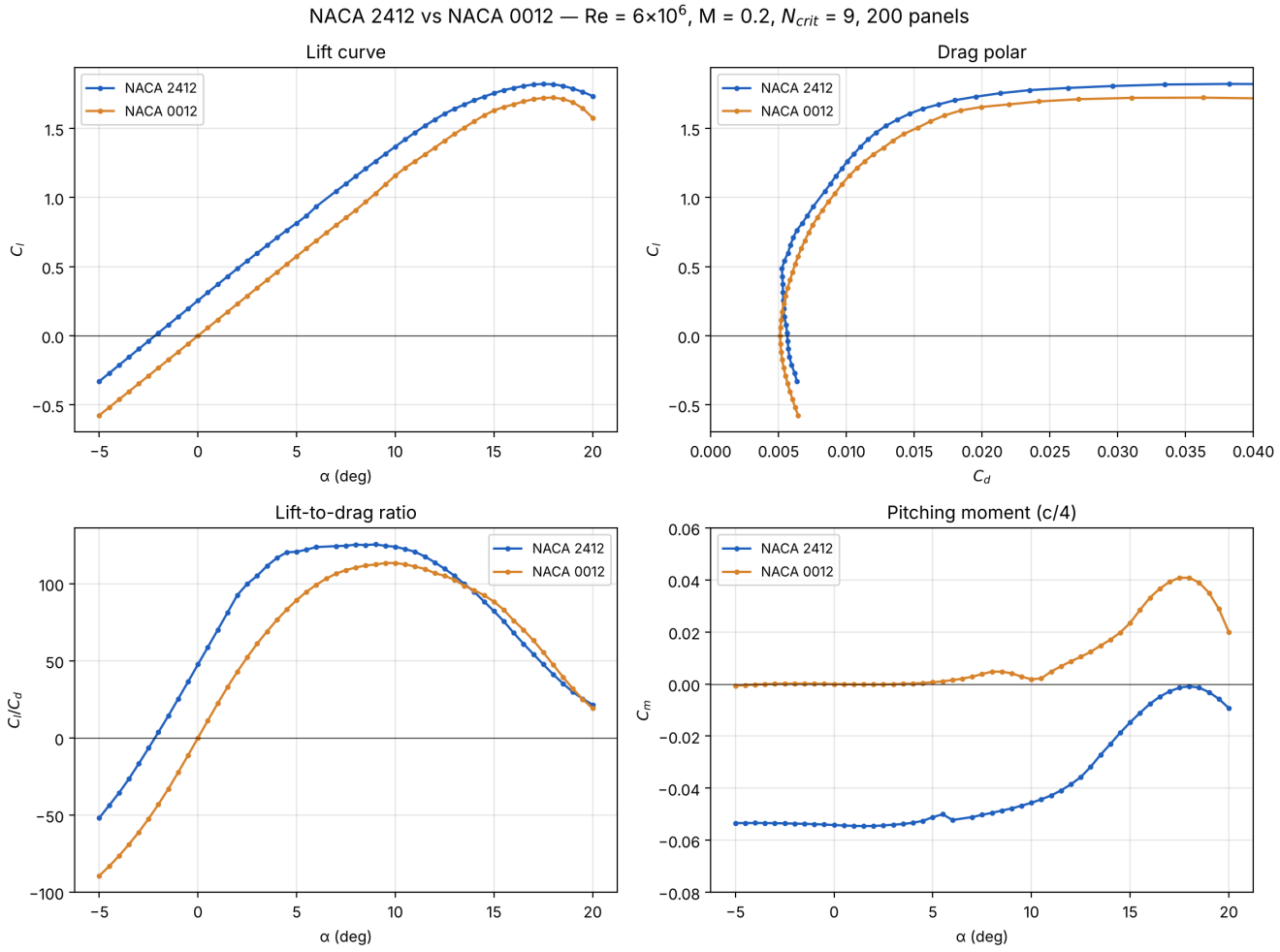



*Figure 3. NACA 2412 vs NACA 0012 at Re = 6 million, M = 0.2, Ncrit = 9.*

| Quantity, Re = 6M | NACA 2412 | NACA 0012 | Effect of camber |
|---|---|---|---|
| Zero-lift angle (deg) | −2.2 | 0.0 | curve shifted left |
| Cl at α = 0° | 0.255 | 0.000 | +0.25 |
| Lift slope (per degree) | 0.114 | 0.115 | unchanged |
| Cl,max | 1.82 | 1.72 | +0.10 |

| Quantity, Re = 6M | NACA 2412 | NACA 0012 | Effect of camber |
|---|---|---|---|
| Stall angle (deg) | 17.5 | 18.0 | 0.5° earlier |
| Minimum Cd | 0.0052 | 0.0051 | about the same |
| Cd at Cl = 0.6 | 0.0057 | 0.0065 | 13% less drag |
| Best Cl/Cd | 126 at 9.0° | 113 at 9.5° | +11% |
| Cm, c/4 | −0.053 | 0.000 | nose-down moment |

*Table 3. Key values at Re = 6 million.*

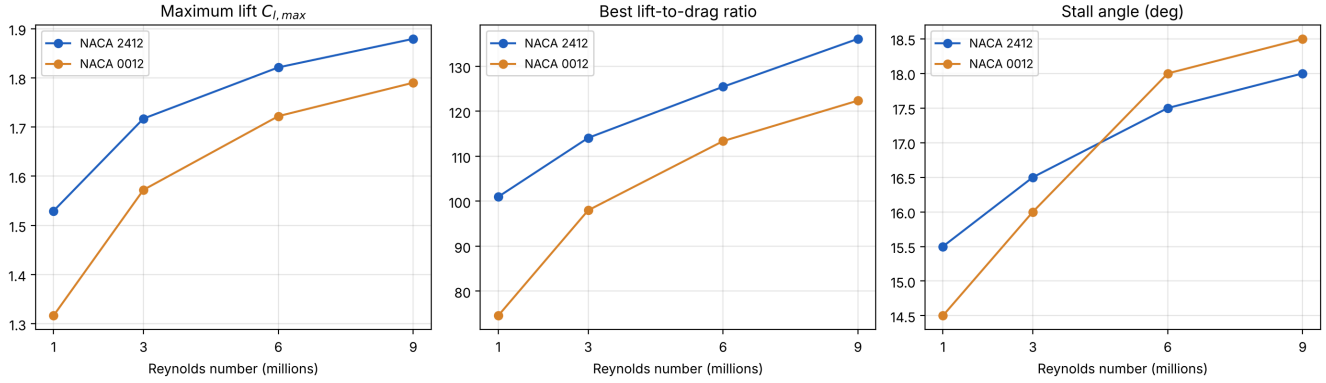



*Figure 4. Maximum lift, best lift-to-drag ratio and stall angle for both airfoils across Reynolds number.*

## 6.1 Lift and moment

The symmetric NACA 0012 gives zero lift at 0°, and its results mirror exactly: Cl = 0.575 at +5° and −0.575 at −5°. Camber moves the zero-lift angle to −2.2° but leaves the slope unchanged, as thin-airfoil theory predicts. The 0012 has Cm ≈ 0 because its aerodynamic centre and centre of pressure both sit at the quarter chord. The camber of the 2412 adds a constant nose-down moment of −0.053, which a real aircraft must balance with its tail.

## 6.2 Drag and efficiency

Both airfoils have the same minimum drag (about 0.0051 to 0.0052), since that is set mostly by thickness and skin friction. The difference is where the minimum sits: at Cl ≈ 0 for the 0012, and at Cl ≈ 0.3 to 0.5 for the 2412, which is where an aircraft actually cruises. At a useful lift of Cl = 0.6 the 2412 has 13% less drag at Re = 6 million, and 28% less at Re = 1 million. As a result its best Cl/Cd is 11 to 35% higher across the Reynolds range.

### 6.3 Maximum lift and stall

The 2412 reaches about 0.1 higher Cl,max at every Reynolds number, and the gap widens to 0.21 at Re = 1 million. The 0012 needs a higher angle to reach its peak and, above Re = 3 million, stalls slightly later. Low Reynolds number hurts the symmetric section more: from 9 to 1 million its best Cl/Cd falls by 39%, against 26% for the 2412.

### 6.4 What this means

Camber buys more lift and better efficiency at cruise for the price of a nose-down pitching moment. That is why the cambered NACA 2412 suits a light aircraft wing, while symmetric sections like the NACA 0012 are used where lift must work equally in both directions: tails, fins, rudders and aerobatic aircraft.

## 7. Limitations

- XFoil is a 2D method: it cannot capture 3D wing effects such as tip vortices and induced drag.
- Post-stall results are unreliable because XFoil does not model large separated regions accurately.
- The surface is assumed perfectly smooth, so drag is optimistic for a real wing.
- Some operating points did not converge and were left as gaps.

## 8. Conclusions

1. The NACA 2412 has a lift slope of 0.114 per degree and a zero-lift angle of −2.2°, both nearly independent of Reynolds number.
2. Increasing Reynolds number from 1 to 9 million raises Cl,max from 1.53 to 1.88, delays stall from 15.5° to 18°, and raises the best Cl/Cd from 101 to 136.
3. Panel resolution matters: 35 panels gave a distorted leading edge and Cp curve, while 200 panels gave smooth, consistent results.
4. Pressure integration reproduced the reported Cl to within 1%, verifying the Cp output.
5. Compared with wind-tunnel data, XFLR5 is accurate in the linear range but overpredicts Cl,max by about 7% and underpredicts minimum drag by about 13%.

Compared with the symmetric NACA 0012, the 2% camber of the NACA 2412 shifts the zero-lift angle from 0° to −2.2°, raises Cl,max by about 0.1, and cuts drag at Cl = 0.6 by 13% at Re = 6 million. The trade-off is a nose-down moment the NACA 0012 does not have, which is why symmetric sections are used for tails and fins.

## 9. Future work

- Extend the comparison to further airfoils.
- Study the effect of Ncrit (freestream turbulence) and forced transition (surface roughness) on drag.
- Use the 2D polars in XFLR5's wing module to analyse a finite 3D wing.
- Compare against a CFD solver for the post-stall region.

## References

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